Supplementary Material: Warmup Is an Update Budget

Scope of the Supplement

This supplement contains nonessential diagnostics and artifact notes. The main submission is self-contained; reviewers need not read this document to evaluate the thesis, mechanism, protocol, or principal results.

1. Additional Probabilistic Summaries

Figure 1 aggregates expected calibration error, negative log-likelihood, and Brier score across the dataset/model groups reported in the main paper. These descriptive metrics follow the same pattern as accuracy: unsafe first-step conditions degrade predictive quality, whereas rescue stays close to corrected-order training.

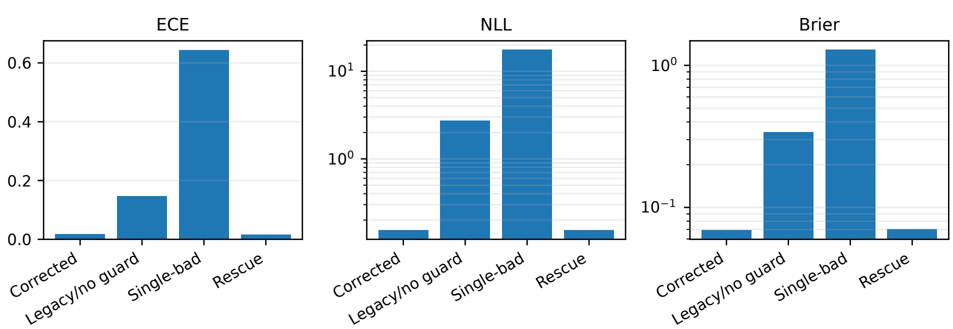

Figure 1: Aggregated descriptive probabilistic metrics for trained conditions. NLL and Brier score are plotted on a log scale. These summaries are descriptive and are not used as inferential tests.

2. Seed-Paired Accuracy Diagnostics

Figure 2 shows the shared-seed trajectories used to summarize paired differences. Lines connect the same seed across conditions within each dataset/model group. The plot is included as a diagnostic for effect direction rather than as a substitute for a larger-seed statistical study.

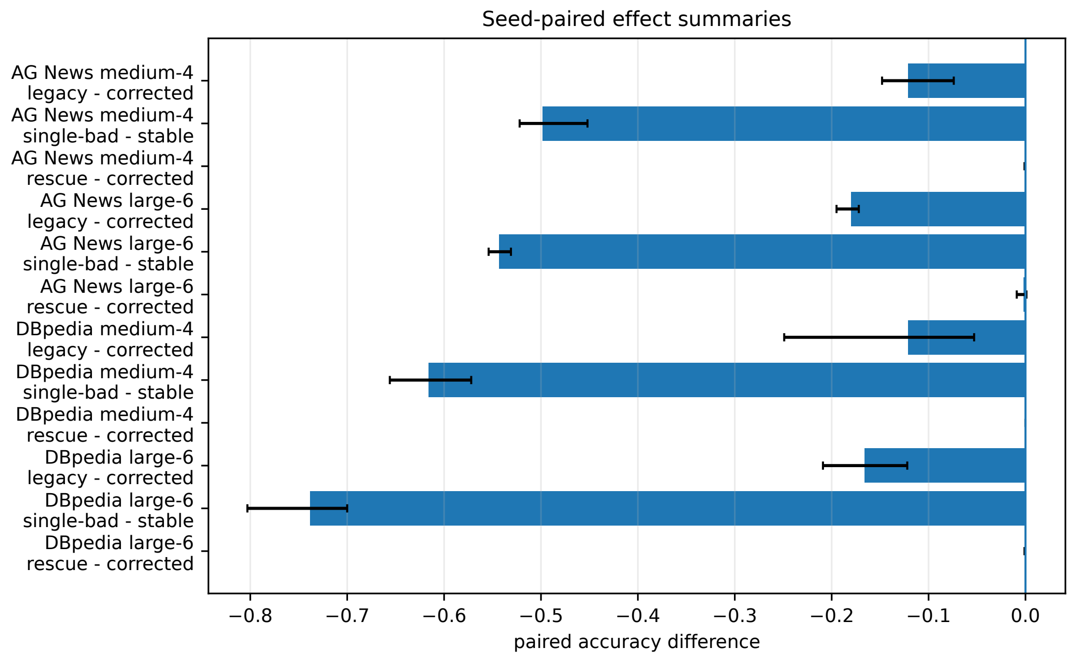

Figure 2: Seed-paired accuracy trajectories. Each line connects the same random seed across trained conditions within a dataset/model group.

3. Toy First-Step Adam Scaling

The toy calculation in Figure 3 uses a clipped unit-norm gradient and Adam epsilon 10^{-9}. It illustrates the proportional dependence of the first update-to-parameter ratio on the actual first-step learning rate in a fixed pre-step state.

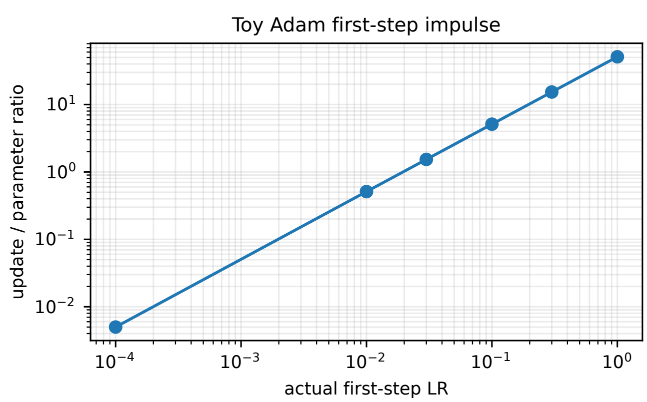

Figure 3: Toy first-step Adam update ratio as a function of the actual first-step learning rate. This is an algebraic diagnostic, not an additional training experiment.

Table 1: Toy first-step Adam scaling values corresponding to Figure 3.

Actual first-step LR	Update norm	Parameter norm	Ratio
1.0000	31.623	0.618	51.128
0.3000	9.487	0.618	15.338
0.1000	3.162	0.618	5.113
0.0300	0.949	0.618	1.534
0.0100	0.316	0.618	0.511
0.0001	0.003	0.618	0.005

4. Guard Actions

Table 2 records guard actions averaged across models and seeds. Fail-fast is a detector and therefore stops; rescue detects the same violation and replaces the unsafe update.

Table 2: Guard action rates averaged across models and seeds.

Dataset	Condition	Trigger	Stop	Rescue
AG News	Legacy/no guard	0.0	0.0	0.0
AG News	Fail-fast	1.0	1.0	0.0
AG News	Rescue	1.0	0.0	1.0
DBpedia	Legacy/no guard	0.0	0.0	0.0
DBpedia	Fail-fast	1.0	1.0	0.0
DBpedia	Rescue	1.0	0.0	1.0

5. Artifact Status

The anonymized source package contains reconstructed summary CSVs and plotting scripts so that figures and tables in the main paper can be rebuilt. It does not include the original raw run directories. A stronger artifact release should add raw per-seed logs, full scheduler traces for every run, guard-event logs, exact manifests, software/hardware versions, and the training-code commit hash.